Asymptotic and structural results for OEIS sequence A112509

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Abstract

For a binary string s of length n, let $f(s)$ be the number of distinct non-negative integers represented by contiguous substrings of s, read in binary with leading zeros suppressed. Define $a(n) = \max_{|s|=n} f(s)$ (OEIS A112509 [1]). The present paper provides the first asymptotic results for this sequence.

We prove that for all $n \geq 64$,

$$a(n) \;\geq\; \tfrac{1}{2}n^2 - 15\,n^{3/2},$$

and hence $\lim_{n\to\infty} a(n)/n^2 = \frac{1}{2}$, matching the elementary upper bound $a(n) \leq n(n+1)/2$. The proof is completely explicit. Beyond the limit, we prove: (i) near-optimal strings can be chosen with ones-density $1-O(n^{-1/2})$; (ii) every optimal string has ones-density $1-O(n^{-1/4})$; (iii) the number of zeros z in any string s reduces the maximum possible $f(s)$ by at least $z(z+1)/2$; (iv) fragmenting the zeros into m runs imposes an additional penalty of at least $\binom{m}{2}$. Computational data for $n \leq 150$ and certified lower bounds through $n = 2 \times 10^9$ are discussed [2].

1 Introduction

1.1 The sequence

For $s \in \{0,1\}^n$, let $f(s)$ denote the number of distinct non-negative integers whose binary representations appear as contiguous substrings of s (leading zeros suppressed: `01` $= 1$, `00` $= 0$). Set $a(n) = \max_{|s|=n} f(s)$. This is OEIS A112509 [1], beginning

$$1, 3, 5, 7, 10, 13, 17, 22, 27, 33, 40, 47, 55, 64, 73, 83, 94, 106, 118, 131, \ldots$$

1.2 Numerical evidence

Exact values for $n \leq 150$ were computed in [2] by exhaustive structured search, and certified lower bounds obtained for much larger n via suffix-array-guided local search. Table 1 records selected values.

n	$a(n)$ or lower bound	$a(n)/n^2 \geq$
50	899	0.3596
100	3,875	0.3875
150	9,070	0.4031
10^6	498,396,662,200	0.4984
10^9	499,943,911,277,037,650	0.49994
2×10^9	1,999,841,302,297,432,600	0.49996

Table 1: Selected values and lower bounds for $a(n)$.

1.3 Main results

Theorem 1 (Limit)**.** $\lim_{n\to\infty} \frac{a(n)}{n^2} = \frac{1}{2}$.

Theorem 2 (Quantitative lower bound)**.** *For every* $n \geq 64$,

$$a(n) \;\geq\; \frac{1}{2}n^2 - 15\,n^{3/2}.$$

Proposition 3 (Density of ones)**.**

(i) *There exists an explicit family of strings* s_n *with* $f(s_n) \geq \frac{1}{2}n^2 - 15n^{3/2}$ *and* $N_0(s_n) = O(\sqrt{n})$, *so the ones-density satisfies* $N_1(s_n)/n = 1 - O(n^{-1/2})$.

(ii) *Every optimal string* s_n^* *satisfies* $N_0(s_n^*) = O(n^{3/4})$, *equivalently* $N_1(s_n^*)/n = 1 - O(n^{-1/4})$.

Section 3 gives the upper bound. Sections 4–6 prove Theorems 1–2. Section 7 proves Proposition 3. Sections 8–9 develop the structural bounds.

2 Definitions

If $w = w_1 \cdots w_k \in \{0,1\}^k$, its binary value is $\nu(w) = \sum_{i=1}^{k} w_i\, 2^{k-i}$. The value set of $s \in \{0,1\}^n$ is $V(s) = \{\nu(s_i \cdots s_j) : 1 \leq i \leq j \leq n\}$, and $f(s) = |V(s)|$. Write $N_1(s)$ and $N_0(s)$ for the number of ones and zeros in s.

3 Upper bound

Lemma 4 (Upper bound)**.** *For every* $n \geq 1$, $a(n) \leq n(n+1)/2$, *and hence* $\limsup_{n\to\infty} a(n)/n^2 \leq 1/2$.

Proof. Every distinct positive integer in $V(s)$ has a unique standard binary representation beginning with `1`, so it is represented by at least one substring of s beginning with `1`. Different positive integers require different such substrings.

If s contains at least one zero then $0 \in V(s)$, represented only by substrings beginning with `0`. At least one of the $n(n+1)/2$ substrings begins with `0`, so the number of leading-`1` substrings is at most $n(n+1)/2 - 1$, giving $f(s) \leq n(n+1)/2$. If s is all ones then $f(s)$ counts distinct positive values, at most $n(n+1)/2$. □

Remark 5. The bound is tight at $n = 2$: `10` gives $V(\texttt{10}) = \{0,1,2\}$, so $a(2) = 3 = 2 \cdot 3/2$.

4 The construction

4.1 Strictly decreasing partitions

Lemma 6 (Decreasing partition)**.** *Let* $k \geq 2$ *and* $M \geq k(k+1)/2$. *Then there exist strictly decreasing positive integers* $b_1 > b_2 > \cdots > b_k \geq 1$ *with* $\sum_{i=1}^{k} b_i = M$. *Moreover, they may be chosen with* $|b_i - M/k| \leq k$ *for all* i.

Proof. Strategy. Consider the arithmetic sequence $c_i = q + k - i$ $(i = 1, \ldots, k)$, which decreases by 1 at each step. Its sum is

$$\sum_{i=1}^{k} c_i = kq + \sum_{i=1}^{k}(k-i) = kq + \frac{k(k-1)}{2}. \tag{1}$$

Choose q so this base sum is as large as possible without exceeding M, and let $r = M - \sum c_i$ be the shortfall. We then add 1 to the first r terms to account for the shortfall.

Definitions. Set

$$q = \Big\lfloor \frac{M}{k} - \frac{k-1}{2} \Big\rfloor, \qquad r = M - kq - \frac{k(k-1)}{2},$$

and define

$$b_i = \begin{cases} q + k - i + 1, & 1 \le i \le r, \\ q + k - i, & r < i \le k. \end{cases}$$

Step 1: $0 \le r < k$. From the definition of q as a floor, $q \le M/k - (k-1)/2 < q + 1$. Multiplying through by k gives

$$kq \;\le\; M - \tfrac{k(k-1)}{2} \;<\; k(q+1),$$

so $0 \le M - kq - k(k-1)/2 < k$, i.e. $0 \le r < k$.

Step 2: Strict decrease. We check all three cases for consecutive pairs $b_i > b_{i+1}$.

- *Both in the first case* ($i < r$, so $i+1 \le r$ as well): $b_i - b_{i+1} = (q+k-i+1) - (q+k-(i+1)+1) = 1 > 0$.
- *Transition* ($i = r \ge 1$, so $i + 1 > r$): $b_r = q + k - r + 1$ and $b_{r+1} = q + k - (r+1)$, giving $b_r - b_{r+1} = q + k - r + 1 - (q + k - r - 1) = 2 > 0$.
- *Both in the second case* ($i > r$): $b_i - b_{i+1} = (q + k - i) - (q + k - i - 1) = 1 > 0$.

Hence $b_1 > b_2 > \cdots > b_k$.

Step 3: The sum equals M. Split the sum at the boundary between the two cases:

$$\begin{aligned} \sum_{i=1}^{k} b_i &= \sum_{i=1}^{r} (q + k - i + 1) + \sum_{i=r+1}^{k} (q + k - i) \\ &= \sum_{i=1}^{k} (q + k - i) + \sum_{i=1}^{r} 1 \\ &= kq + \frac{k(k-1)}{2} + r \;=\; M, \end{aligned}$$

where the last equality uses the definition of r and (1).

Step 4: All terms are positive. From $M \ge k(k+1)/2$ we get $M/k \ge (k+1)/2$, so

$$\frac{M}{k} - \frac{k-1}{2} \;\ge\; \frac{k+1}{2} - \frac{k-1}{2} = 1,$$

hence $q = \lfloor M/k - (k-1)/2 \rfloor \ge 1$. Since $r < k$, the index $i = k$ falls in the second case, giving $b_k = q + k - k = q \ge 1$.

Step 5: Proximity bound. Every first-case term satisfies $q < q + k - r + 1 \le b_i \le q + k$, and every second-case term satisfies $q \le b_i = q + k - i \le q + k - r - 1 < q + k$. So all $b_i \in [q,\, q + k]$. From the floor definition, $M/k \in [q + (k-1)/2,\; q + (k+1)/2)$, and the width of $[q, q + k]$ around any point in this interval is at most $(k+1)/2 < k$ in each direction. Hence $|b_i - M/k| < k$ for all i. □

4.2 Parameters and string

Fix $n \ge 64$ and set

$$k = \Big\lfloor \tfrac{\sqrt{n}}{4} \Big\rfloor, \qquad M = n - (k-1). \tag{2}$$

Since $n \ge 64$ we have $k = \lfloor \sqrt{n}/4 \rfloor \ge \lfloor 8/4 \rfloor = 2$.

Why $M \ge n/2$. Since $k \le \sqrt{n}/4$, we have $\sqrt{n}/4 \le n/2$ (equivalently $\sqrt{n} \le 2n$, true for all $n \ge 1$), so $k - 1 < k \le \sqrt{n}/4 \le n/2$. Hence $M = n - (k-1) > n - n/2 = n/2$.

Why $n/2 \geq k(k+1)/2$. This is equivalent to $n \geq k(k+1)$. Since $k \leq \sqrt{n}/4$ we have

$$k(k+1) \leq k \cdot 2k = 2k^2 \leq 2 \cdot \frac{n}{16} = \frac{n}{8} \leq n,$$

where the second inequality uses $k \leq \sqrt{n}/4$ so $k^2 \leq n/16$. Hence $M \geq n/2 \geq k(k+1)/2$, and Lemma 6 applies.

Choose $b_1 > b_2 > \cdots > b_k \geq 1$ with $\sum b_i = M$ and $|b_i - M/k| \leq k$, as in Lemma 6. Define

$$s = \underbrace{1^{b_1}}_{\text{block } 1} \; 0 \; \underbrace{1^{b_2}}_{\text{block } 2} \; 0 \cdots 0 \; \underbrace{1^{b_k}}_{\text{block } k}, \tag{3}$$

consisting of k one-blocks separated by $k-1$ single zeros. Its length is $\sum b_i + (k-1) = M + (k-1) = n$.

5 An injective family of substrings

For integers i, j, α, β satisfying

$$1 \leq i, \quad j \geq i+2, \quad j \leq k, \quad 1 \leq \alpha \leq b_i, \quad 1 \leq \beta \leq b_j, \tag{4}$$

define $u_{i,j,\alpha,\beta}$ to be the substring of s beginning α positions before the end of block i and ending after the first β ones of block j. In binary,

$$u_{i,j,\alpha,\beta} = 1^{\alpha}\, 0\, 1^{b_{i+1}}\, 0\, 1^{b_{i+2}}\, 0 \cdots 1^{b_{j-1}}\, 0\, 1^{\beta}. \tag{5}$$

The condition $j \geq i+2$ ensures at least one full interior block $(b_{i+1}, \ldots, b_{j-1})$. Every $u_{i,j,\alpha,\beta}$ starts and ends with 1, so it is already a standard binary expansion of a positive integer.

Lemma 7 (Injectivity)**.** *The map* $(i, j, \alpha, \beta) \mapsto u_{i,j,\alpha,\beta}$ *is injective on the range* (4)*.*

Proof. Suppose $u_{i,j,\alpha,\beta} = u_{i',j',\alpha',\beta'}$ as binary words. The run-length encoding (RLE) of (5), listing alternating run-lengths of ones then zeros, is

$$(\alpha,\ 1,\ b_{i+1},\ 1,\ b_{i+2},\ 1,\ \ldots,\ b_{j-1},\ 1,\ \beta).$$

Matching entries gives $\alpha = \alpha'$, $\beta = \beta'$, and $(b_{i+1}, \ldots, b_{j-1}) = (b_{i'+1}, \ldots, b_{j'-1})$. Since $b_1 > \cdots > b_k$ is strictly decreasing, every consecutive subsequence occurs in exactly one position, so $i = i'$ and $j = j'$. □

6 Quantitative lower bound and the limit

Proof of Theorem 2. By Lemma 7, the family $\{u_{i,j,\alpha,\beta}\}$ consists of pairwise distinct positive integers, so

$$a(n) \geq f(s) \geq \sum_{\substack{1 \leq i < j \leq k \\ j \geq i+2}} b_i b_j. \tag{6}$$

Expanding:

$$\sum_{j \geq i+2} b_i b_j = \frac{(\sum b_i)^2 - \sum b_i^2}{2} - \sum_{i=1}^{k-1} b_i b_{i+1}. \tag{7}$$

Step 1. Since $\sum b_i = M = n - k + 1$ and $k \leq \sqrt{n}/4$,

$$\frac{M^2}{2} \geq \frac{(n - \sqrt{n}/4)^2}{2} \geq \frac{n^2}{2} - \frac{n^{3/2}}{4}.$$

Step 2. Write $b_i = M/k + d_i$ with $|d_i| \le k$. Then $\sum b_i^2 \le M^2/k + k^3$. Since $k \ge \sqrt{n}/8$ (as $\lfloor x \rfloor \ge x/2$ for $x \ge 2$), $M^2/k \le n^2/(\sqrt{n}/8) = 8n^{3/2}$, and $k^3 \le (\sqrt{n}/4)^3 = n^{3/2}/64$, so

$$\sum_{i=1}^{k} b_i^2 < 9n^{3/2}. \tag{8}$$

Step 3. By AM–GM, $b_i b_{i+1} \le (b_i^2 + b_{i+1}^2)/2$, so

$$\sum_{i=1}^{k-1} b_i b_{i+1} \le \sum_{i=1}^{k} b_i^2 < 9n^{3/2}. \tag{9}$$

Conclusion. Substituting into (7):

$$a(n) \ge \frac{n^2}{2} - \frac{n^{3/2}}{4} - \frac{9n^{3/2}}{2} - 9n^{3/2} = \frac{n^2}{2} - \frac{55}{4}n^{3/2} > \frac{n^2}{2} - 15n^{3/2}. \qquad \square$$

Proof of Theorem 1. Lemma 4 gives $\limsup a(n)/n^2 \le 1/2$. Theorem 2 gives $a(n)/n^2 \ge 1/2 - 15/\sqrt{n} \to 1/2$. $\square$

Corollary 8. *For all $n \ge 64$, $n^2/2 - 15n^{3/2} \le a(n) \le n(n+1)/2$, so $a(n)/n^2 = 1/2 + O(n^{-1/2})$.*

Remark 9. The lower bound $n^2/2 - 15n^{3/2}$ is negative for $n < 900$ (since $n^2/2 > 15n^{3/2}$ iff $\sqrt{n} > 30$). For such n the inequality in Theorem 2 is trivially satisfied by the non-negativity of $a(n)$, and carries no information. The bound is meaningful only for $n \ge 900$; for small n the exact values from computational search (Table 1) give much sharper information.

7 Density of ones

7.1 Near-optimal strings

Proof of Proposition 3(i). The string $s = s_n$ from (3) has $f(s_n) \ge n^2/2 - 15n^{3/2}$ by Theorem 2, and contains exactly $k - 1 \le \sqrt{n}/4$ zeros, so $N_0(s_n) = O(\sqrt{n})$ and $N_1(s_n)/n = 1 - O(n^{-1/2})$. $\square$

7.2 Zero-position penalty

Lemma 10 (Zero-position bound)**.** *Let $s \in \{0,1\}^n$ have zeros at positions $p_1 < \cdots < p_z$. Then*

$$f(s) \le \frac{n(n+1)}{2} - \sum_{r=1}^{z} (n - p_r + 1).$$

In particular, $f(s) \le n(n+1)/2 - z(z+1)/2$.

Proof. For each $x \in V(s)$ choose a minimum-length representative w_x. The map $x \mapsto w_x$ is injective. If $x > 0$ then w_x starts with `1` (otherwise removing leading zeros gives a shorter representative). So distinct positive values correspond to distinct leading-`1` substrings.

Substrings starting at zero-position p_r number $n - p_r + 1$ and none starts with `1`, giving the first inequality. The second follows by placing zeros as far right as possible ($p_r = n - z + r$), which minimises $\sum(n - p_r + 1) = z(z+1)/2$. $\square$

7.3 Optimal strings have few zeros

Proof of Proposition 3(ii). Let s_n^* be optimal with $z_n = N_0(s_n^*)$. By Theorem 2 and Lemma 10,

$$\frac{n^2}{2} - 15n^{3/2} \le a(n) = f(s_n^*) \le \frac{n(n+1)}{2} - \frac{z_n(z_n+1)}{2}.$$

Hence $z_n(z_n+1)/2 \le 15n^{3/2} + n/2 + 1$, giving $z_n = O(n^{3/4})$ and $N_1(s_n^*)/n = 1 - O(n^{-1/4})$. $\square$

8 Upper bounds via the zero structure

Lemma 10 treats all zeros equally. We now refine the bound using the *run structure* of the zeros.

Let the maximal zero-runs of s have lengths $c_1, \ldots, c_m$, with the i-th run beginning at position a_i and $r_i = n - (a_i + c_i) + 1$ positions to its right.

Lemma 11 (Run-sensitive bound)**.** $f(s) \le n(n+1)/2 - \sum_{i=1}^{m}\big(c_i r_i + \binom{c_i+1}{2}\big)$.

Proof. Apply Lemma 10 and group by zero-runs. The i-th run occupies positions $a_i, \ldots, a_i + c_i - 1$, contributing $\sum_{t=0}^{c_i-1}(n - a_i - t + 1) = c_i(n - a_i + 1) - c_i(c_i - 1)/2 = c_i r_i + \binom{c_i+1}{2}$ to the sum. □

9 Fragmentation penalty

Corollary 12 (Fragmentation)**.** *If s has z zeros in m maximal zero-runs, then*

$$f(s) \;\le\; \frac{n(n+1)}{2} - \frac{z(z+1)}{2} - \binom{m}{2}.$$

Proof. It suffices to show $\sum_i\big(c_i r_i + \binom{c_i+1}{2}\big) \ge z(z+1)/2 + \binom{m}{2}$. Since consecutive zero-runs are separated by at least one 1, for $i < m$ we have $r_i \ge \sum_{j>i} c_j + (m - i)$, so $\sum_i c_i r_i \ge \sum_{i<j} c_i c_j + \sum_{i=1}^{m-1} c_i(m - i)$. The identity $\binom{z+1}{2} = \sum_{i<j} c_i c_j + \sum_i (c_i^2 + c_i)/2$ gives $\sum_{i<j} c_i c_j + \sum_i \binom{c_i+1}{2} = z(z+1)/2$. Finally $\sum_{i=1}^{m-1} c_i(m - i) \ge \sum_{i=1}^{m-1}(m - i) = \binom{m}{2}$ (each $c_i \ge 1$). □

Remark 13. Corollary 12 shows that fragmenting the zeros into more runs always reduces the upper bound on $f(s)$, regardless of positions. Each additional run incurs an extra penalty of at least $m - 1$ (the new run number), on top of the baseline $z(z+1)/2$.

10 Computational evidence

Exact values ($n \le 150$). Exact values of $a(n)$ were computed for all $n \le 150$ in [2] using three independent methods: unconstrained Metropolis–Hastings search (at most 10 restarts, reaching the optimum for every $n \le 150$), constrained Metropolis–Hastings with a fixed run-length template, and exact exhaustive template search. On every overlap the methods agreed.

Large-scale certified lower bounds. A suffix-array-guided local search (*surgical nudge*) was used for n up to 2×10^9 [2]. In each iteration the algorithm finds the highest-LCP collision in the suffix array, searches a neighbourhood for a density-preserving swap that resolves it, and accepts only swaps that strictly increase the exact score. Since the score is monotone non-decreasing, every recorded value is a rigorous lower bound. At $n = 2 \times 10^9$ a lower bound of $a(n)/n^2 \ge 0.49996$ was obtained.

Comparison with theory. The ratio $a(n)/n^2$ rises from 0.3596 at $n = 50$ to 0.4031 at $n = 150$, and exceeds 0.49996 at $n = 2 \times 10^9$. Theorem 2 gives $a(n)/n^2 \ge 1/2 - 15/\sqrt{n}$, which reaches 0.4875 only at $n = 14{,}400$. The computational lower bounds substantially exceed the theoretical bound at moderate n, confirming that the constant 15 in $15n^{3/2}$ is far from tight.

Entropy of optimal solutions. Let S_n denote the set of all strings achieving $a(n)$ and $H(n) = \log_2 |S_n|$. For $n \le 150$, $H(n)$ is almost always zero or very small: the typical optimum is unique or nearly so, with $H(n) \le 10.4$ throughout. Optimal strings share long common prefixes; the zero-separator structure is completely rigid in 28 cases and varies by at most one separator in a further 119 cases out of $n \le 150$.

11 Heuristic remarks

To maximise $f(s)$, one seeks to maximise the number of leading-`1` substrings that remain pairwise distinct after leading zeros are stripped. Long repeated patterns create large common-prefix collisions, inflating LCP values and causing many pairs of substrings to agree. The explicit construction in Section 4 avoids this by using *strictly decreasing* one-block lengths: any two substrings $u_{i,j,\alpha,\beta}$ and $u_{i',j',\alpha',\beta'}$ spanning different block-pairs $(i,j) \neq (i',j')$ have different interior block-length sequences, making them provably distinct (Lemma 7).

Empirically, all optimal strings for $n \leq 150$ follow a similar principle: they begin with a long run of ones, followed by decreasing-length one-blocks with a fixed separator prefix $1, 2, 1, 1, \ldots$, and have ones-density growing toward 1. This structure is consistent with the proven density bounds, though the actual empirical zero count at small n grows far more slowly than the theoretical $O(n^{3/4})$ bound would suggest.

12 A worked example

Example 14 ($n = 169$, $k = 3$)**.** With $n = 169$: $k = \lfloor 13/4 \rfloor = 3$, $M = 169 - 2 = 167$. Choose

$$b_1 = 57, \quad b_2 = 56, \quad b_3 = 54; \qquad 57 > 56 > 54 \geq 1, \quad 57 + 56 + 54 = 167\,.$$

The construction gives the 169-bit string

$$s = \underbrace{1^{57}}_{\text{block 1}}\ 0\ \underbrace{1^{56}}_{\text{block 2}}\ 0\ \underbrace{1^{54}}_{\text{block 3}}\,,$$

of length $57 + 1 + 56 + 1 + 54 = 169$.

The only valid pair with $j \geq i + 2$ and $j \leq k = 3$ is $(i,j) = (1,3)$, with interior block $b_2 = 56$. For each (α, β) with $1 \leq \alpha \leq 57$ and $1 \leq \beta \leq 54$, the substring $u_{1,3,\alpha,\beta} = 1^\alpha\, 0\, 1^{56}\, 0\, 1^\beta$. By Lemma 7 these $57 \times 54 = 3{,}078$ substrings are pairwise distinct positive integers, so $a(169) \geq 3{,}078$.

Note: the explicit bound $n^2/2 - 15n^{3/2} < 0$ at $n = 169$, reflecting that the constant 15 is conservative and the asymptotic result requires large n to dominate.

Acknowledgements

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References

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